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Multiple pathways of restorative responses to environmental exposures during urban walking: Evidence from a crossover field trial

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ABSTRACT

Nature-based walking, as an effective non-pharmacological health-promoting intervention, is associated with restorativeness. However, the influence of multiple physical environmental factors on restorative responses remains insufficiently understood under real-world, equivalent contexts. This randomized crossover field experiment examined how pedestrian-level physical environmental variability during walking relates to subjective and physiological restorative responses. Thirty-six participants completed walking sessions in all combinations of two seasons and two environmental contexts (green-urban and dense-urban). Wearable sensors continuously recorded thermal, acoustic, and air pollution parameters as well as physiological signals, with perceived restorativeness and affect assessed at multiple experimental stages. Partial least squares structural equation modeling (PLS-SEM) and linear mixed-effects models were applied jointly to examine exposure–response pathways.

Results demonstrate that heterogeneous environmental exposures within the same urban walking context elicit differentiated short-term restorative responses through multiple parallel exposure–response pathways. Affective and physiological components showed partial independence, with no significant mediation effects. Specifically, trafficand pollution-related stressors, e.g., $PM_{2.5}$ ($p = 0.007$) and O_3 ($p = 0.029$), were primarily associated with electrodermal arousal. Perceptual and experiential factors, e.g., visual satisfaction ($p < 0.001$) and ventilation ($p = 0.045$), predominantly influenced positive and negative affect. A significant interaction between dense-urban walking and $PM_{2.5}$ ($p = 0.015$) further indicated context-dependent modulation of electrodermal responses. Scale-dependent effects were observed, with thermal conditions operating at broader contextual levels and traffic-related stressors acting more directly at the pedestrian scale. Together, these findings highlight the importance of incorporating exposure management into urban design.

1. Introduction

With continued global urban expansion and population growth (United Nations, 2024), urban living environments are facing increasing pressures and challenges. These trends have increasingly emerged as major public health concerns, accompanied by a rising burden of mental health disorders documented in global health reports and epidemiological studies (GBD 2019 Mental Disorders Collaborators, 2022; World Health Organization, 2024). Extensive research has established links between health outcomes and urban living environments, which support physical activity and social interaction (Mazumdar et al., 2018; Sallis et al., 2016), thereby influencing residents' daily experiences and mental health (Bratman et al., 2019; Pandya et al., 2025), including those arising from routine mobility and outdoor walking environments (Frank et al., 2019).

Attention Restoration Theory and Stress Recovery Theory provide complementary perspectives on the mental benefits of urban natural spaces. Attention Restoration Theory emphasizes recovery from

Abbreviations: Rh, Relative Humidity; *V*, Wind Velocity; *Ta*, Air Temperature; *Tmrt*, Mean Radiant Temperature; UTCI, Universal Thermal Climate Index; PLS-SEM, Partial Least Squares Structural Equation Modeling; LMM, Linear Mixed-effects Model; AIC, Akaike Information Criterion; EDA, Electrodermal Activity; SCL, Skin Conductance Level; SCR, Skin Conductance Response; *Tsk*, Skin Temperature; Leq, Equivalent Continuous Sound Pressure Level; PANAS, Positive and Negative Affect Schedule; PRS, Perceived Restorativeness Scale; PRND, Participant's Residential Neighborhood Density.

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attentional fatigue through nature exposure (Kaplan and Kaplan, 1989; Kaplan, 1995), whereas Stress Recovery Theory highlights physiological stress reduction and affective responses elicited by non-threatening natural environments (Ulrich, 1983; Ulrich et al., 1991). Despite these different emphases, both theories converge on the assumption that nature exposure produces restorative benefits, which have been robustly supported by empirical evidence (Bratman et al., 2019, Markevych et al., 2017). While these frameworks primarily focus on environmental features, they also incorporate behavioral factors such as physical activity, providing a foundation for nature-based interventions (Hartig et al., 2014). Nevertheless, these frameworks are largely grounded in a simplified urban–nature dichotomy, which limits their ability to account for the dynamic and heterogeneous exposure patterns of real-world urban environments, leaving the underlying mechanisms of restorative responses insufficiently explained.

Walking ranks among the most feasible physical activity modes, characterized by simplicity, self-paced participation, and broad accessibility (Kelly et al., 2018; Morris and Hardman, 1997), and also serves as a key pathway for environmental exposure in daily life. Consequently, it is recognized as a low-cost and scalable public health intervention (World Health Organization, 2020). Walking interventions across varying durations have demonstrated diverse health outcomes, including reductions in stress, depressive symptoms, and cardiovascular risk (Hanson and Jones, 2015, Kelly et al., 2018). During walking, exposure to urban greenery may generate different restorative responses depending on environmental context and exposure duration (Hunter et al., 2019; Tu et al., 2026b). Synthesized evidence consistently supports the mental health benefits of nature-based walking, particularly for anxiety reduction and affective improvement (Chen et al., 2025, Tu et al., 2026b). Walking in greener environments enables the dual benefit of physical activity and nature exposure (Lee et al., 2024; Olafsdottir et al., 2020), thereby producing cumulative health gains through every step.

Walking inherently involves multi-domain, spatiotemporally dynamic environmental exposures along its routes, rather than a simplified urban–nature dichotomy or steady-state environmental contact. These exposures are typically understood as spanning multiple sensory and environmental domains, including visual, thermal, acoustic, and air quality dimensions (Tu et al., 2026b). Built environment characteristics and visual elements—such as urban landscapes, greenery, and streetscape features—have been associated with restorative outcomes, with effects differing across specific attributes (Celikors and Wells, 2022; Luo et al., 2024; Mahia et al., 2025b; Xiang et al., 2021). Beyond visual factors, thermal conditions further modulate restorative responses (Dong et al., 2025; Tu et al., 2026a), and certain acoustic conditions may facilitate restorativeness (Jeon et al., 2021), whereas traffic-related noise and air pollutants tend to attenuate these benefits (Dzhambov et al., 2018; Schaupp et al., 2025). This suggests that restorative responses during walking are not driven solely by natural elements, but emerge from multi-sensory environmental exposures encountered along walking routes (Frank et al., 2019; Wang et al., 2016).

Exposure to walking environments is spatiotemporally heterogeneous across diverse built environments and multiple scales, while being further driven by dynamic and self-determined walking behaviors with variable timing. For example, at the urban scale, larger cities (>100 km^2) exhibit stronger heat island effects and higher thermal stress than smaller towns (<10 km^2) (Chen et al., 2026b), while at the neighborhood level, heat vulnerability varies substantially (Chen et al., 2026a). At the pedestrian scale, buildings and street trees shape sunlight and shade distribution, modifying solar radiation and thermal perception (Lai et al., 2023; Zhao et al., 2026), thereby elevating pedestrians' local skin temperature and stress levels (Mahia et al., 2025a). Similarly, variations in built environment, traffic intensity, and individual walking behavior generate heterogeneous traffic-related air pollutant exposures (Armengol et al., 2024; Dhondt et al., 2012). Collectively, such layered variations highlight the spatiotemporal heterogeneity of walking-related environmental exposures across space, time, and scale, suggesting that restorative experiences unfold within complex and uneven environmental contexts (Frank et al., 2019, Tu et al., 2026b).

Existing research on walking-based restorativeness is paradoxically constrained by overly broad conceptualizations of restorative environments. Many studies predominantly employ simplified environmental typologies (e.g., "green" vs. "urban") to represent restorative contexts, without further differentiation of environmental characteristics within these categories (Tu et al., 2026b). Such simplifications implicitly assume that environmental exposures along a given route remain spatially homogeneous. Although a few recent studies have begun to quantify fine-grained environmental exposures during walking—such as traffic noise (Schaupp et al., 2025), thermal conditions (Song et al., 2024), and visual attributes (Luo et al., 2024, Zhang et al., 2022)—such approaches remain scarce within walking-based restorativeness research, particularly in simultaneous multi-dimensional assessment. In addition, many prior studies rely predominantly on self-reported restorativeness, although objective physiological indicators are increasingly emerging as complementary measures (Kabisch et al., 2021, Tu et al., 2026b). This dichotomy precludes the quantitative characterization of concurrent environmental exposures during walking, leaving their effects on psychological and physiological responses, as well as underlying restorative processes insufficiently understood.

While previous studies have provided valuable insights into individual environmental factors in walking-based restorativeness, most related studies have examined these factors in isolation as covariates or moderators, leaving their joint influences and relative strengths on restorative responses unclear. To date, only a few studies have considered multiple concurrent environmental stressors during walking, such as thermal variables, noise, and air pollution, to examine their relative effects on restorative responses. However, these efforts have generally required mobile sensing approaches to measure multi-exposure environments during walking. Among the limited studies, Lanki et al. (2017) linked cardiovascular indicators with traffic-related stressors (i.e., noise and particulate matter), and Tu et al. (2026a) demonstrated that adverse thermal and air pollution exposures modified restorative responses.

Existing evidence remains largely fragmented, as co-occurring environmental exposures have yet to be analyzed within an integrated framework (Li et al., 2026). At the same time, emerging but limited evidence suggests joint and context-dependent effects on restorative responses. For example, a recent study on static restorative environments showed that neural responses are enhanced by greenery combined with multisensory stimuli—such as soundscapes and volatile organic compounds—through synergistic effects (Li et al., 2026). Similarly, in nature-based walking studies, Song et al. (2024) found that environmental context significantly modulated thermal perception under equivalent exposure conditions. Schaupp et al. (2025) examined the interaction between environmental context and traffic noise in relation to restorative responses, but observed no significant effect. Broadly, empirical evidence on interactions among environmental exposures and context remains scarce.

Overall, key limitations of existing research include: (1) the methodological reliance on simplified environmental categorizations and homogeneous exposure assumptions, (2) the predominant focus on single environmental factors with limited consideration of their interactions and context-dependent effects, and (3) the dependence on subjective measures. As a result, integrated, multi-factor, real-time characterization of heterogeneous environmental exposures during walking, as well as their combined and context-dependent effects on both physiological and perceptual restorative responses, remains lacking. Additionally, a previous literature review indicates that global studies on walking-based restorativeness have been predominantly conducted in temperate regions, while evidence from subtropical and tropical urban contexts remains relatively limited (Tu et al., 2026b). Guangzhou, as a megacity with a population exceeding ten million, is characterized by high-density built environments, serving as a

representative subtropical urban context.

Building on the identified research gaps, this study goes beyond merely confirming the restorative benefits of nature-based walking compared with dense-urban environments. Instead, it adopts a multi-factor environmental exposure framework integrating simultaneous physiological and perceptual measurements to examine relationships between environmental exposures and short-term restorative responses during walking. Accordingly, two research questions are proposed:

(1) Within a single walking session, how do multiple concurrently occurring physical environmental exposures—such as air quality, acoustic conditions, and microclimatic characteristics—relate to short-term restorative responses? How do their effects differ in direction and magnitude?

(2) Within conventional restorative environment categorizations (e.g., green-urban versus dense-urban in this study), do the exposure-response relationships between physical environmental exposures and short-term restorative responses remain consistent across environmental contexts, or do they exhibit potentially interactive, context-dependent differences?

To address these questions, a randomized crossover experiment was conducted across two different walking environments in two seasons. Multiple environmental exposure data were simultaneously collected using a wearable mobile sensing platform and temporally aligned with restorative measures during walking. This integrated design enables synchronous assessment of co-occurring exposure-response relationships within a single walking session under a unified research framework.

2. Methods

2.1. Experimental design

2.1.1. Experimental settings

The experiment was conducted in Guangzhou, China, which has a warm temperate with dry winter and hot summer (Cwa) climate under the Köppen–Geiger climate classification system (Peel et al., 2007). To examine exposure–response relationships within conventional restorative environment categorizations, two walking routes with contrasting environmental characteristics were selected. Both routes were located within 2 km of each other in Guangzhou's central Tianhe District, providing a comparable urban background.

The green-urban route, located on the campus of South China University of Technology, featured abundant vegetation, with half of the route along a lake (Fig. 1a), representing a walking environment with comprehensive natural elements. In contrast, the dense-urban route near Tianhebei Road traversed a high-density residential area and was flanked by roadways. One half of the route was adjacent to a primary arterial road, and the other half to a secondary arterial road (Fig. 1b), representing a typical traffic-influenced urban walking environment.

The experiment was conducted on rain-free days during two separate periods, thereby encompassing a wider range of environmental conditions: from December 13 to December 29, 2024 (winter session), and from April 14 to June 4, 2025 (summer session). Experiments were rescheduled in case of rainfall to ensure consistent outdoor conditions. Differences in seasonal study duration were primarily driven by weather constraints and participant availability under field conditions. All trials were conducted within fixed daytime windows (9:30–11:00 and 14:30–16:30) under relatively moderate solar radiation to avoid excessive thermal stress on participants. These designs accommodated



Fig. 1. a. Route map of the green-urban walking route; b. route map of the dense-urban walking route; c. representative segment of the green-urban route showing blue-space features; d. representative segment of the green-urban route without blue-space features; e. experimental procedure illustrating participants walking with a guide while physical environment measurements were recorded; f. representative segment of the dense-urban route adjacent to a secondary arterial road; g. representative segment of the dense-urban route adjacent to a primary arterial road. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

naturally occurring environmental variation while preserving comparability across repeated measurements within the same environmental contexts in a field-based setting.

2.1.2. Study population

The experiment employed a convenience sampling strategy, recruiting participants through university advertisements and online recruitment. Eligible participants were healthy young adults aged 18–35 years who were free from serious physical or mental health issues and capable of sustaining continuous walking for at least 30 min. Considering the proximity of the walking routes to the university campus, university-based participants were selected to ensure feasibility and facilitate protocol adherence during repeated field sessions, while also reducing inter-individual variability in physiological and perceptual responses during walking to improve the stability of exposure–response estimation. The study protocol was approved by the Ethics Committee of the Second Affiliated Hospital of South China University of Technology (approval number: K-2023-056-01), and complied with the ethical standards for human research. All participants provided written informed consent and received monetary compensation after each session.

A total of 37 participants (19 females, 18 males), mainly graduate students from universities in Guangzhou, were recruited. Baseline health status and physical activity levels were assessed via self-reported questionnaires prior to recruitment. Given prior evidence linking urban land density to mental health (Sui et al., 2022; Xu et al., 2023), participants' residential neighborhood density (PRND; past 3 years) was self-reported and included as a covariate. All participants met the predefined inclusion criteria and were free from physical or mental health conditions. Individuals with professional athletic training were excluded to reduce potential bias in physiological responses. The sample was characterized by generally low levels of smoking and no reported alcohol abuse. Participants were instructed to abstain from alcohol and caffeinated beverages before each session. One participant withdrew during the summer session, while the remaining participants completed all four walking interventions (Fig. 2). Detailed demographics are provided in Table 1.

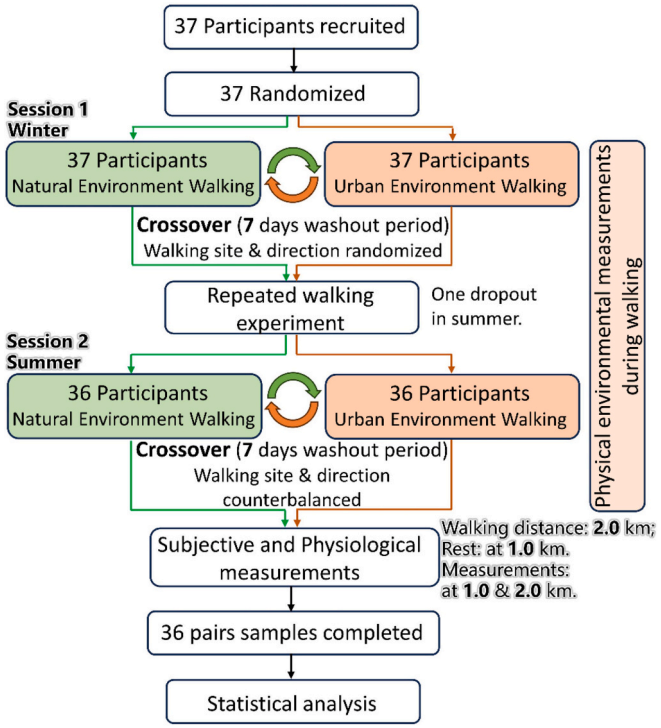


Fig. 2. Study design of the trial.

Table 1
Participant demographic characteristics ($n = 37$).

Variable	Mean (SD)	Variable	N (%)
Age	24.2 (1.9)	Gender (% female)	51.4%
BMI (females)	19.3(2.4)	Alcohol use: Non-abuser (%)	100%
BMI (males)	23.0(2.3)	Smoking status: Non-smoker (%)	94.6%
Main living environment in the past 3 years (%)			
Suburban	8.1%	Low-density urban	48.6%
Medium-density urban	29.7%	High-density urban	13.5%

Note: BMI indicates Body Mass Index.

2.1.3. Experimental protocol

In this crossover trial, participants completed both the green-urban and dense-urban walking routes once in each of the two seasons. For the first session (winter), they were randomly assigned to start with one route, walking 2.0 km at a self-selected pace with guidance, before completing the other route at least one week later (Fig. 2). The second session (summer) followed the same protocol. Allocation sequence and walking direction (clockwise or counterclockwise) were randomized to balance potential order effects across repeated measurements and identical route-direction combinations. A one-week interval between routes, commonly adopted as a washout period in crossover studies, together with the seasonal interval between sessions, was intended to reduce potential carryover and familiarity effects. No more than two participants were arranged to walk together. Prior to each walking session, participants were reminded by the experimenter to avoid non-essential conversation and were briefly informed about the procedures.

The 2.0 km walking distance was based on previous review suggesting that nature-based walking exposures of approximately 1.5–2.5 km are associated with relatively effective psychological benefits (Tu et al., 2026b), while restorative outcomes in young adults have been reported to peak around a 2.0 km exposure distance (Tu et al., 2026a). The intermediate observation points at 1.0 km (resting points in Fig. 1), corresponding to the predefined boundary between environmentally heterogeneous route segments, were incorporated to assess restorative responses contrasts across walking environments. The walking pace was self-selected without control to preserve ecological validity and reflect real-world walking conditions. Walking time was recorded, with a mean pace of 14.151 ± 1.191 min per km in this study.

The procedure followed a pretest-posttest design identical across environments (Fig. 3). Upon arrival, participants entered a resting stage (T1), during which physiological devices were attached and baseline health (SF-12) and subjective measures were collected. This was followed by a stress induction stage (T2), in which participants completed two 5-min tasks, including reading and Sudoku. The two tasks involved different cognitive processes with moderate difficulty, and task versions varied across sessions. This stage was designed to induce mild cognitive load prior to walking, following established protocols in restoration research (Stevenson et al., 2018). The second round of subjective measures was administered immediately afterward (T3).

Participants then began the walking intervention. During the first half (T4), they walked 1.0 km along the assigned route. This was followed by an intermediate transition stage (T5), consisting of a 5-min break for restorative response assessment, including the third round of subjective measures and brief hydration, before the second walking segment. During the second half (T6), participants completed the remaining 1.0 km. Physiological and environmental data were continuously recorded during T4 and T6, and subjective satisfaction ratings were collected at designated points. The post-walk stage (T7) included the final subjective measures and removal of all monitoring devices. The rest and transition stages (T1 and T5) were conducted at designated shaded locations along each walking route, where participants remained stationary.

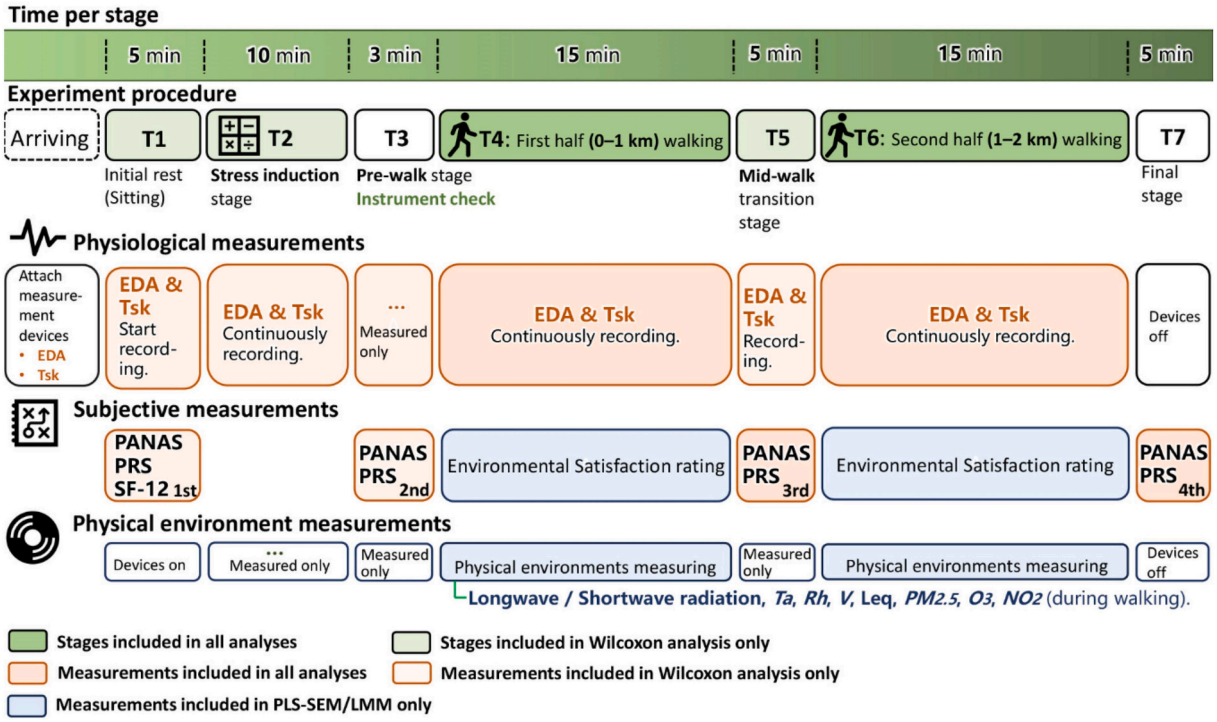


Fig. 3. Experimental procedure. (Note: Subjective questionnaires administered at T3, T5, and T7 were analyzed as T2-, T4-, and T6-stage responses, respectively, to align with the corresponding environmental and physiological measurements.)

2.2. Measurements

2.2.1. Subjective measurements

Subjective restorative responses were repeatedly assessed across four stages during each walking intervention: the resting stage (T1), post-stress induction (T2), after the first half of walking (T4), and after the second half of walking (T6). Although the analytic subjective measures were administered at T3, T5, and T7, they assessed responses to the immediately preceding primary analytical stages and were therefore labeled as T2-, T4-, and T6-stage measures to align with the corresponding environmental and physiological data. Perceived restorativeness was measured using the Perceived Restorativeness Scale (PRS; 22 items, 7-point Likert scale) (Hartig et al., 1997). Affective dimensions were measured using the Positive and Negative Affect Schedule (PANAS; 20 items, 5-point Likert scale) (Watson et al., 1988). To minimize participant burden associated with repeated assessments, only these two primary instruments were administered across the walking stages (42 items in total, approximately 2 min). In addition, baseline general health status was assessed once at resting stage (T1) using the 12-Item Short Form Survey (SF-12) (Ware et al., 1996).

During walking, perceived environmental quality—including thermal, visual, and acoustic satisfaction—was assessed at fixed locations along the routes (5-point Likert scale). Assessments were collected at approximately 100 m intervals (19 assessment points per route), with locations generally aligned with pedestrian crossings or route landmarks. At each location, participants briefly paused under experimenter guidance to complete the assessment ratings, which typically required approximately 10 s.

2.2.2. Physiological measurements

Physiological responses included skin temperature (*Tsk*) and electrodermal activity (EDA), continuously recording from participants' arrival until the completion of the entire experiment. Skin temperature was recorded with iButton temperature loggers (DS1922L, Maxim Integrated), sampled at 30-s intervals. The loggers were attached to four body locations—neck, right scapula, left hand, and right shin—following the ISO 9886 standard. Mean skin temperature was calculated as a weighted average of these four sites, with weighting coefficients of 0.28 each for the neck, scapula, and shin, and 0.16 for the hand (summing to 1.00) (International Organization for Standardization, 2004).

Electrodermal activity, commonly indexed by skin conductance to assess emotional changes and sympathetic arousal (Kreibig, 2010; Veeranki et al., 2026), consists of a tonic component—skin conductance level (SCL)—and a phasic component—skin conductance responses (SCRs) (Greco et al., 2016). Owing to their established validity and reliability, previous walking-related restorativeness studies have widely employed both SCL and SCR (Luo et al., 2024, Tu et al., 2026a, Zhang et al., 2022). Skin conductance was recorded using a wearable electrode system (Shimmer 3 GSR+, Shimmer) to derive EDA measures during walking. Following clinical validation, the device was shown to highly accurate measurements (Gatti et al., 2018). Electrodes were placed on participants' wrists, and data were recorded at 51.2 Hz.

2.2.3. Physical environmental measurements

Physical environmental indicators during walking were measured using the same backpack-mounted monitoring system as in our previous experiments (Tu et al., 2026a). This system recorded radiation, wind velocity (*V*), air temperature (*Ta*), relative humidity (*Rh*), sound levels, and air pollutants. The integrated system included multiple sensors and instruments, with detailed specifications, measurement ranges, and accuracy provided in Appendix A, Table A.1. All sensors were operated and calibrated according to manufacturer specifications. The same devices were used across sessions, and device synchronization and functionality were checked prior to each experimental session.

During the walking stages, the backpack was carried by an experimenter, who followed the participant at a distance of 2–3 m to minimize visual interference (Fig. 1e). Shortwave and longwave radiation were recorded at 10-s intervals to calculate mean radiant temperature (*Tmrt*, °C). These parameters, together with synchronously measured *Ta*, *Rh*, and *V* (sampled every 10 s), were used to compute the Universal Thermal Climate Index (UTCI), a widely adopted comprehensive indicator of

human thermal comfort (Jendritzky et al., 2012). During walking, air pollutants—including $PM_{2.5}$, NO_2, and O_3—were continuously monitored (sampled every 60 s), and the equivalent continuous sound pressure level (Leq) was recorded by a handheld sound level meter.

2.3. Statistical analysis

2.3.1. Electrodermal activity signal analysis

Ledalab was used to analyze the calibrated skin conductance signal, which reflected electrodermal activity (Benedek and Kaernbach, 2010). Following Ledalab recommended preprocessing procedures and the suggestions of Greco et al. (2016), a first-order low-pass Butterworth filter with a 5 Hz cutoff was applied to reduce high-frequency noise and motion-related artifacts prior to signal decomposition. The data were downsampled to 32 Hz to limit signal fluctuations, smooth the overall waveform, and reduce computational load while preserving relevant electrodermal activity signal characteristics, following the example preprocessing settings provided in Ledalab. Continuous deconvolution analysis allowed the decomposition of the signal into tonic (mean SCL) and phasic components (mean SCR amplitude and SCR counts) (Greco et al., 2016). In addition, SCR rate was defined as the average number of significant SCRs per minute, which reflected stimulus-evoked activity.

Across the two experimental sessions, one participant dropped out of the summer session (146 completed observations). During EDA signal quality and stability assessment, one participant exhibited extreme outliers and substantial signal instability across sessions; therefore, these EDA observations were invalidated and treated as invalid. Additionally, skin conductance signals from three other participants were not successfully recorded for one walking intervention each due to experimental issues. In total, seven EDA observations were considered missing in subsequent analyses.

2.3.2. Universal thermal climate index calculation

Mean radiant temperature (*Tmrt*, °C) was calculated using measured longwave (*Li*) and shortwave (*Ki*) radiation flux densities in six directions (upward, downward, front, back, left, and right). As shown in Eq. (1), angular weighting factors ($Wi = 0.06$ for upward and downward sensors; 0.22 for front, back, left, and right sensors) and absorption coefficients ($\alpha l = 0.97$ for longwave radiation and $\alpha k = 0.70$ for shortwave radiation) were applied to compute *Tmrt* for an upright reference pedestrian (Höppe, 1992).

A logarithmic wind profile was applied to convert the wind velocity measured at the sensor height (1.9 m) to the standard height of 10 m (V_{10}), as required for the UTCI calculation. In this conversion, $Z = 1.9$ m represented the height of the wind velocity sensor, $Z_R = 10$ m represented the reference height, and $Z_0 = 0.01$ m represented the roughness length, as detailed in Eq. (2) (Havenith et al., 2012).

$$T_{mrt} = \sqrt[4]{\left[\sum_{i=1}^{6} w_i(a_i K_i + a_i L_I)\right] \Big/ (a_i \sigma)} - 273.15\mathrm{K}(^{\circ}\mathrm{C}) \quad (1)$$

$$V_{10m} = V_{1.9m} \times log(Z/Z_0) \times log(Z_R/Z_0)^{-1} \quad (2)$$

UTCI values were derived from *Tmrt* along with corresponding *Ta*, *Rh*, and V_{10} using the pythermalcomfort Python package (Tartarini and Schiavon, 2020).

2.3.3. Analytical methods

Data analysis was conducted in RStudio (R Core Team, 2025) and SmartPLS 4 (Ringle et al., 2024). (1) Stage-level differences (T1–T6) within each experiment were assessed using the Wilcoxon signed-rank test for paired samples, separately for season and walking environment comparisons. The approach provides robust, distribution-free inference for ordinal and non-Gaussian data in field-based settings (Hollander et al., 2013). To control for false positives from multiple comparisons, *p*-values were additionally adjusted using the Benjamini–Hochberg false discovery rate procedure ($\alpha = 0.05$), which is suitable for numerous correlated comparisons (Benjamini and Hochberg, 1995).

(2) Partial Least Squares Structural Equation Modeling (PLS-SEM) was utilized as an exploratory analytical framework to disentangle the intervention-based pathways of restorative change. To evaluate multi-exposure restorative dynamics, PLS-SEM enables the simultaneous assessment of effect direction and magnitude within a unified multivariate analytical framework, as well as the identification of key exposure-response pathways and interaction-relevant structures for further analysis. (3) Linear mixed-effects models (LMMs) were then applied to cross-validate the exposure-response pathways identified in PLS-SEM, and to further examine interaction effects between walking environment type and environmental indicators on restorative responses (i.e., T4 and T6), using the "lme4" package (Bates et al., 2015).

As a subclass of structural equation modeling (SEM), PLS-SEM uses ordinary least squares regression to maximize the explained variance of target endogenous latent variables (Hair et al., 2019). Rather than validating predefined theoretical models, it is particularly suited to exploring and identifying underlying variable structures within complex field-based systems. In the present study, PLS-SEM was applied to disentangle the multivariate relationships among environmental exposures and multilevel restorative responses within real-world walking contexts, reflecting the assumption of multi-factor environmental effects. Methodologically, the method demonstrates robustness under conditions of non-normality and limited sample sizes, under which it can provide stable parameter estimates and valid significance testing (Hair et al., 2012). Although frequently used in secondary data, PLS-SEM is also applicable to controlled experimental and intervention-based designs (Singleton et al., 2023).

2.3.4. Statistical power

The minimum sample size for the within-subject (paired) design was calculated using G*Power 3.1 (Faul et al., 2009). Assuming a medium effect size (Cohen's $d = 0.50$), $\alpha = 0.05$, and power = 0.85, the calculation indicated that 32 participants were required. A total of 37 participants were recruited, with one withdrawal. For PLS-SEM, the inverse square root method was applied (Hair Jr et al., 2021; Kock and Hadaya, 2018), and path significance was assessed via bootstrapping with 10,000 resamples. Assuming $\alpha = 0.05$ and a minimum path coefficient of 0.25, the estimated minimum sample size was 99.

Each completed experimental session was treated as an independent observation, yielding 146 observations for analysis. All questionnaire and environmental variables were complete, whereas seven EDA observations were identified as missing or invalid. Missing EDA values were imputed using the mean values of the corresponding experimental condition prior to subsequent analyses. Analyses requiring complete paired data were conducted using 144 paired observations due to the withdrawal of one participant, whereas other analyses retained all 146 observations.

2.3.5. PLS-SEM specification and model validation

In the PLS-SEM, physical environmental variables were modeled as exogenous single-indicator constructs, whereas subjective and physiological responses were specified as endogenous reflective latent constructs. Analyses were based on within-walk T6–T4 pairwise difference values, with physical environmental exposures and restorative response variables represented as between-segment contrasts within each walking session to ensure observation independence and isolate intervention-related effects. Given the moderate correlation between O_3 and NO_2 (Spearman's $\rho = 0.526$, $p < 0.001$), only one representative gaseous pollutant indicator of was retained in the primary model to maintain model parsimony. To better reflect the acute walking exposure investigated in this study, O_3 was retained in the primary PLS-SEM model, consistent with the shorter averaging exposure time commonly adopted for O_3 than for NO_2 in current epidemiological evidence

(Orellano et al., 2020; World Health Organization, 2021). In addition, sensitivity analyses using NO_2 in place of O_3 were conducted to assess the robustness of the results (Appendix Table C.4).

The structural model included three reflective latent constructs, each measured by two indicators: Positive Subjective Outcomes (PRS and PANAS-Positive), Electrodermal Arousal (SCL and SCR amplitude), and Perceived Environmental Quality (environmental satisfaction rating). Conceptually independent outcomes (Thermal Satisfaction, SCR rate, and PANAS-Negative) were modeled as single-indicator reflective constructs. Skin Temperature, Thermal Satisfaction, and Perceived Environmental Quality were specified as endogenous intermediary constructs linking physical environmental exposures to restorative responses. Individual-level covariates (age, BMI, gender, and PRND) were included as single-indicator exogenous control variables with direct paths to endogenous constructs.

Confirmatory assessments were performed for multi-indicator latent variables to evaluate measurement model quality, including *outer loadings*, *composite reliability* (CR), *average variance extracted* (AVE), and the *heterotrait-monotrait ratio* (HTMT). These metrics were not applicable to formative indicators, which were all single-indicator formative constructs in this study.

2.3.6. LMM specification and model evaluation

Given that the PLS-SEM examined the effects of between-segment environmental contrasts during walking (T6–T4), which may compress variation in absolute exposure levels, LMMs were further used to cross-validate significant pathways identified in the PLS-SEM. Stage-level repeated observations during walking (T4 and T6) were used in the LMMs, with restorative responses calculated as baseline-corrected changes relative to T2 (i.e., T4–T2 and T6–T2) to isolate within-person restorative changes during walking. In the LMMs, participant identity was specified as a random intercept to control for repeated measurements. Environment type (green-urban vs. dense-urban) was treated as a fixed factor. Individual-level covariates and cumulative walking duration were included as control variables. Restorative response variables were subjected to inverse normal transformation followed by z-score standardization prior to model estimation to satisfy distributional assumptions regarding random effects and residuals (Laird and Ware, 1982).

Consistent with the study objective of identifying the physical environmental factors underlying restorative responses, season was not included as a primary covariate in the main models. This was primarily because season is inherently associated with multiple physical environmental exposures, and adjustment for season could potentially alter the interpretation of their independent effects. Therefore, season-adjusted models were used as sensitivity analyses to evaluate the robustness of the findings (Appendix C). Diagnostic residual and Q–Q plots are provided in Appendix D. VIF diagnostics indicated low multicollinearity among predictors in both the PLS-SEM and LMM models (maximum VIF = 4.337).

For each dependent variable, a stepwise modeling strategy was applied, including: (1) a baseline model with environment type, walking duration, and individual-level covariates; (2) models adding predictors identified as significant or near-significant in the PLS-SEM; (3) a combined main-effects model; and (4) a full interaction model testing predefined Dense-urban walking × Predictor terms, as well as identified interactions among physical environmental indicators. Model performance was evaluated using Akaike Information Criterion (AIC) and marginal R^2 to assess improvements in explanatory power.

3. Results

3.1. Descriptive statistics

Descriptive statistics for physical environmental variables and subjective satisfaction during walking (T4 and T6) are presented in Table 2.

Table 2
Summary of physical environmental measures and subjective satisfaction during walking across experiment sessions (T4 + T6).

Variable Mean (± SD)	Winter		Summer	
	Green-urban	Dense-urban	Green-urban	Dense-urban
Physical Environmental Measures				
UTCI (°C)	13.660 ± 2.817	15.189 ± 2.429	28.281 ± 2.478	30.896 ± 3.010
Ta(°C)	17.020 ± 2.243	18.177 ± 2.183	27.538 ± 2.525	29.668 ± 2.290
Rh (%)	38.007 ± 10.322	42.466 ± 9.583	58.036 ± 15.185	62.657 ± 11.671
V (m/s)	1.416 ± 0.241	1.179 ± 0.136	1.218 ± 0. 173	1.086 ± 0.136
Leq (dBA)	57.459 ± 1.311	65.936 ± 1.086	59.328 ± 1.164	67.108 ± 1.133
$PM_{2.5}$ (mg/m^3)	0.001 ± 0.001	0.003 ± 0.001	0.001 ± 0.001	0.000 ± 0.001
O_3 (ppm)	0.053 ± 0.008	0.067 ± 0.007	0.082 ± 0.016	0.076 ± 0.017
Perceived Environmental Satisfaction Rating				
Visual Sat.	3.684 ± 0.367	3.116 ± 0.417	3.571 ± 0.495	3.088 ± 0.365
Acoustic Sat.	3.342 ± 0.397	2.764 ± 0.402	3.251 ± 0.505	2.709 ± 0.429
Thermal Sat.	4.023 ± 0.749	4.232 ± 0.706	3.581 ± 0.706	3.570 ± 0.886

Within-condition stage differences (|T6 − T4|) are reported in Appendix A, Table A.2. Further, Table 3 presents descriptive statistics for restorative responses—including subjective and physiological measures—across the walking interventions (T4 and T6), and 95% confidence intervals were provided in Appendix A, Table A.3.

The SF-12 questionnaire was administered once at T1 in each experimental session to assess baseline health status. No significant differences were observed between sessions (all p-values = 0.269–0.967), indicating stable physical and psychological health conditions throughout the study period. The five PANAS items with the highest mean scores for each affective dimension (Positive and Negative Affect, 10 items respectively) were then presented within each condition and stage, arranged according to the circumplex model of affect proposed by Russell (1980), with spatial positions based on Schaupp et al. (2025) (Fig. 5a–h); detailed statistics are in Appendix A. A Tables A.4 and A.5.

3.2. Results of the Wilcoxon signed-rank tests

3.2.1. Subjective measures

PRS: No significant changes were observed during stress induction (T2) across seasons or environments (all $p > 0.05$). After walking, green-urban routes showed significant PRS increases relative to baseline (T6–T2) in both winter and summer ($p < 0.001$; $p = 0.039$), whereas no effects were detected at midpoint (T4–T2; all $p > 0.05$). In dense-urban settings, winter walking produced significant improvements at both T4 and T6 (both $p < 0.001$). Summer dense-urban walking showed a significant increase only at midpoint (T4–T2: $p = 0.007$), with no end-stage effect (T6–T2: $p = 0.072$). Across seasons, PRS was consistently higher in green-urban than dense-urban environments at both T4 and T6 (all $p \leq 0.008$) (Fig. 4a).

PANAS-Positive: During stress induction (T2), positive affect decreased significantly in all conditions except summer green-urban walking ($p = 0.609$). After walking onset, only winter dense-urban walking showed a significant increase at midpoint (T4–T2: $p = 0.037$), whereas other conditions exhibited no significant changes (all $p > 0.05$). At T6, positive affect was highest under winter green-urban walking and was significantly higher than both winter and summer dense-urban conditions ($p = 0.005$; $p < 0.001$) (Fig. 4b).

At the item level, *Interested* consistently ranked among the top

Table 3
Physiological and subjective measures across experimental stages.

Variable Mean (± SD)	Winter				Summer			
	Green-urban		Dense-urban		Green-urban		Dense-urban	
	T4 (1 km)	T6 (2 km)	T4 (1 km)	T6 (2 km)	T4 (1 km)	T6 (2 km)	T4 (1 km)	T6 (2 km)
PRS	4.756 ± 0.794	4.997 ± 0.806	4.228 ± 1.148	3.841 ± 1.066	4.587 ± 0.835	4.722 ± 0.856	4.058 ± 0.969	3.885 ± 0.889
PANAS-Positive	30.028 ± 5.398	31.278 ± 5.952	28.639 ± 6.567	28.444 ± 6.500	28.500 ± 6.423	29.028 ± 7.263	27.667 ± 6.254	27.694 ± 7.110
PANAS-Negative	15.750 ± 6.207	14.583 ± 6.282	15.611 ± 6.460	15.528 ± 6.287	15.306 ± 4.818	14.889 ± 5.049	16.000 ± 5.442	15.778 ± 5.740
Mean SCL (μS)	4.334 ± 3.160	5.668 ± 3.766	3.235 ± 2.359	4.575 ± 3.101	2.739 ± 1.924	4.124 ± 2.584	2.602 ± 2.375	3.912 ± 3.252
Mean SCR amplitude (μS)	0.497 ± 0.540	0.594 ± 0.638	0.634 ± 0.655	0.601 ± 0.555	0.365 ± 0.260	0.514 ± 0.382	0.390 ± 0.419	0.483 ± 0.433
SCR rate	13.921 ± 1.744	13.704 ± 2.268	14.011 ± 2.303	13.902 ± 1.949	14.777 ± 2.337	14.737 ± 1.946	14.685 ± 1.892	15.374 ± 1.528
Tsk (°C)	30.817 ± 1.302	30.901 ± 1.290	31.380 ± 1.092	31.430 ± 1.087	33.445 ± 0.985	33.669 ± 0.803	33.715 ± 0.914	33.661 ± 0.804
Baseline Status:								
Cumulative Time (min)	14.820 ± 1.461	14.201 ± 0.889	13.689 ± 0.939	13.553 ± 0.412	14.506 ± 1.257	14.534 ± 1.099	13.773 ± 1.092	14.163 ± 1.392
SF-12(T1)	42.752 ± 6.842		43.316 ± 7.616		44.444 ± 7.240		43.446 ± 7.618	

Note: T4 = first half of the walking stage (1 km); T6 = second half of the walking stage (remaining 1 km). PRS range: 1–7; PANAS range: 10–50; SF-12 range: 0–100.

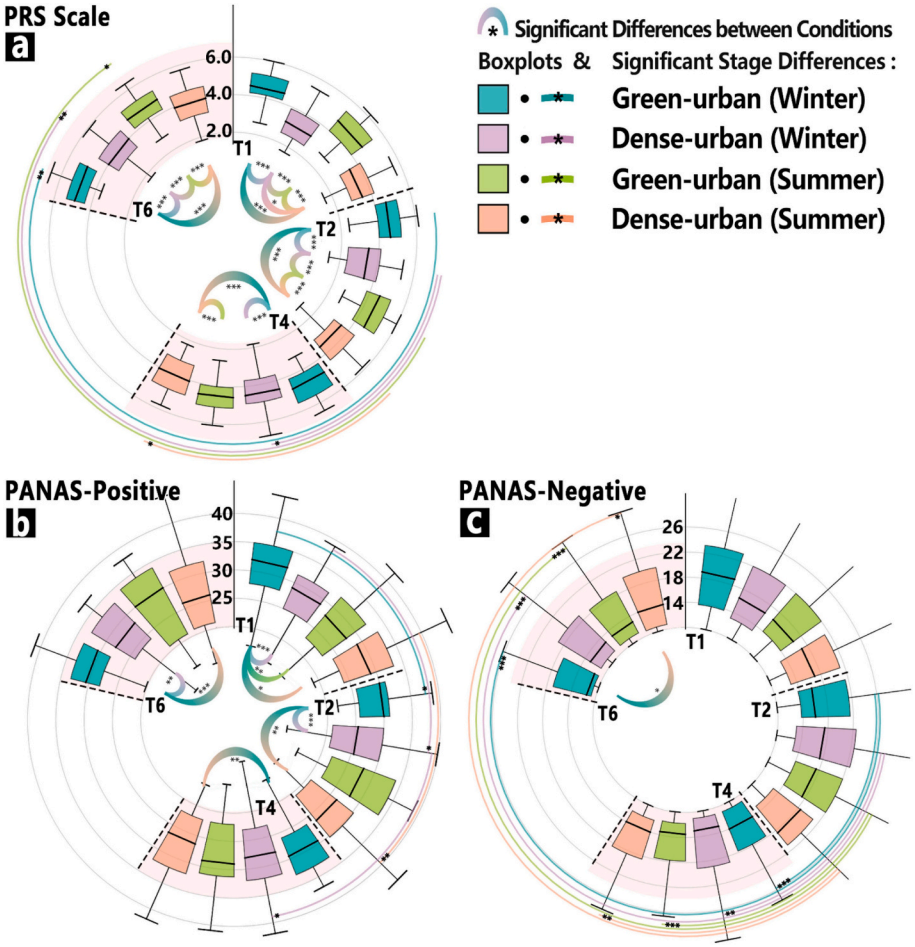


Fig. 4. Boxplots of stage-level restorative responses of subjective measures with Wilcoxon test results across experiments and stages: a. Perceived Restorativeness Scale; b. PANAS (Positive and Negative Affect Schedule)-Positive; c. PANAS-Negative. (Note: Thin curved lines outside the boxplots indicate statistically significant pairwise differences between stages within the same experiment. Thick gradient-colored arcs within the central circle indicate statistically significant differences between experiments at the same stage. Connected groups indicate significant pairwise comparisons, whereas the absence of connecting arcs indicates non-significant between-group comparisons. Significance levels are denoted as $p < 0.05$: *, $p < 0.01$: **, $p < 0.001$: ***.)

positions across stages, with *Active* and *Strong* more frequently advancing into leading ranks from T2 to T6 (Fig. 5a–h). The remaining items (e.g., *Excited*, *Attentive*) generally stayed within the top five across conditions. The only exception was winter green-urban walking, where



Fig. 5. Distribution of the top-rated PANAS items before and after the walking stage in the circumplex model of affect: a. Winter Green-urban (T2); b. Winter Green-urban (T6); c. Winter Dense-urban (T2); d. Winter Dense-urban (T6); e. Summer Green-urban (T2); f. Summer Green-urban (T6); g. Summer Dense-urban (T2); h. Summer Dense-urban (T6). T2 represents the baseline stress induction stage and T6 the post-walking stage. From the 20 PANAS items (10 Positive Affect and 10 Negative Affect items), the five items with the highest mean scores were selected separately for each affective dimension within each condition and stage. Items are arranged following circumplex model of affect proposed by Russell (1980). Spatial positioning of PANAS items is based on Schaupp et al. (2025). Values represent mean ratings on a 5-point Likert scale (1–5). Green = Positive Affect; orange = Negative Affect; darker shades indicate higher mean scores. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

high-arousal, moderately pleasant items—including *Excited*, *Enthusiastic*, and *Active*—showed a noticeable increase (Fig. 5b). Collectively, green-urban walking showed observable but non-significant improvements in positive affect, with a stable configuration of core items.

PANAS-Negative: No significant changes were observed during stress induction (T2), and no baseline differences were detected between conditions (all $p > 0.05$). Walking was associated with significant reductions in negative affect at both T4 and T6 across all environments and seasons (all $p < 0.05$). Between-condition comparisons further showed lower negative affect after winter green-urban walking (T6) compared with summer dense-urban walking ($p = 0.048$).

At the item level, *Distressed*, *Upset*, and *Nervous* consistently ranked highest at T2 and remained prominent at T6 across all trial conditions, indicating a stable composition of negative affect before and after walking (Fig. 5a–h). The low-arousal, highly unpleasant item *Distressed* was particularly salient following dense-urban walking in both seasons (Fig. 5d & h). In contrast, high-arousal negative items—such as *Nervous* and *Jittery*—as well as moderately unpleasant items including *Upset* and *Ashamed*, showed notable reductions after walking. Overall, walking exerted a robust and consistent effect in reducing negative affect regardless of condition, particularly among high-arousal items (Fig. 4c).

3.2.2. Physiological measures

SCL: Significant stage-to-stage increases were observed across all stages and conditions (all $p < 0.001$). At baseline (T2), SCL was highest in the winter green-urban condition compared with all others (all $p \leq$

0.001). During walking, SCL in winter green-urban remained significantly higher than all other conditions at T4 (all $p < 0.001$) and higher than both winter and summer dense-urban conditions at T6 ($p = 0.020$; $p = 0.005$). In summer, green-urban walking showed higher SCL than dense-urban only at T4 ($p = 0.048$), with no difference at T6 (Fig. 6a).

SCR amplitude: SCR amplitude did not change significantly during stress induction (T2–T1; all $p > 0.100$), but increased significantly during walking across all conditions (T4–T2 and T6–T2; all $p < 0.001$). At baseline (T2), the summer dense-urban condition showed significantly lower SCR amplitude than all other conditions (all $p < 0.05$). During walking, only winter dense-urban walking exhibited higher SCR amplitude at T4 compared with both summer walking conditions (all $p < 0.05$). No other between-condition differences were observed (Fig. 6b).

SCR Rate: Except for winter dense-urban walking ($p = 0.004$), SCR rate showed no significant change during stress induction (T2–T1; all $p > 0.500$). At baseline (T2), SCR rate was lower in winter green-urban than winter dense-urban ($p = 0.010$). Walking induced significant SCR rate increases across all conditions (T4–T2 and T6–T2; all $p < 0.001$). At T4, winter green-urban showed lower SCR rates than both summer walking conditions (all $p < 0.050$). At T6, SCR rates were higher in summer than in winter across both environments (all $p < 0.010$), while no within-season environmental differences were observed (Fig. 6c).

3.2.3. Interpretation of Wilcoxon signed-rank tests

Taken together, green-urban walking generally produced more favorable restorative responses, with winter green-urban walking yielding the highest PRS and the lowest Negative affect at the end of the walking stages (T6). Positive affect did not increase consistently across conditions but remained relatively higher after winter green-urban walking. Notably, a greater presence of high-arousal, moderately pleasant items was observed under this condition.

For physiological measures, winter green-urban walking was characterized by higher tonic SCL and relatively lower phasic SCR activity at T6, accompanied by higher PRS and PANAS-Positive scores and lower PANAS-Negative scores. These concurrent changes indicate elevated physiological arousal alongside improved perceived restorativeness and positive affect, and reduced negative affect. At the midpoint of winter dense-urban walking (T4), participants exhibited partial short-term restorative responses even in highly urbanized environments: PRS and positive affect increased, negative affect decreased, and SCL and SCR amplitudes were higher.

3.3. Partial least squares structural equation modeling analysis

3.3.1. Measurement model assessment

The results of the measurement model evaluation are presented in Table 4. Indicator reliability was evaluated using *outer loadings*, with values above 0.708 explaining >50% of the variance and considered

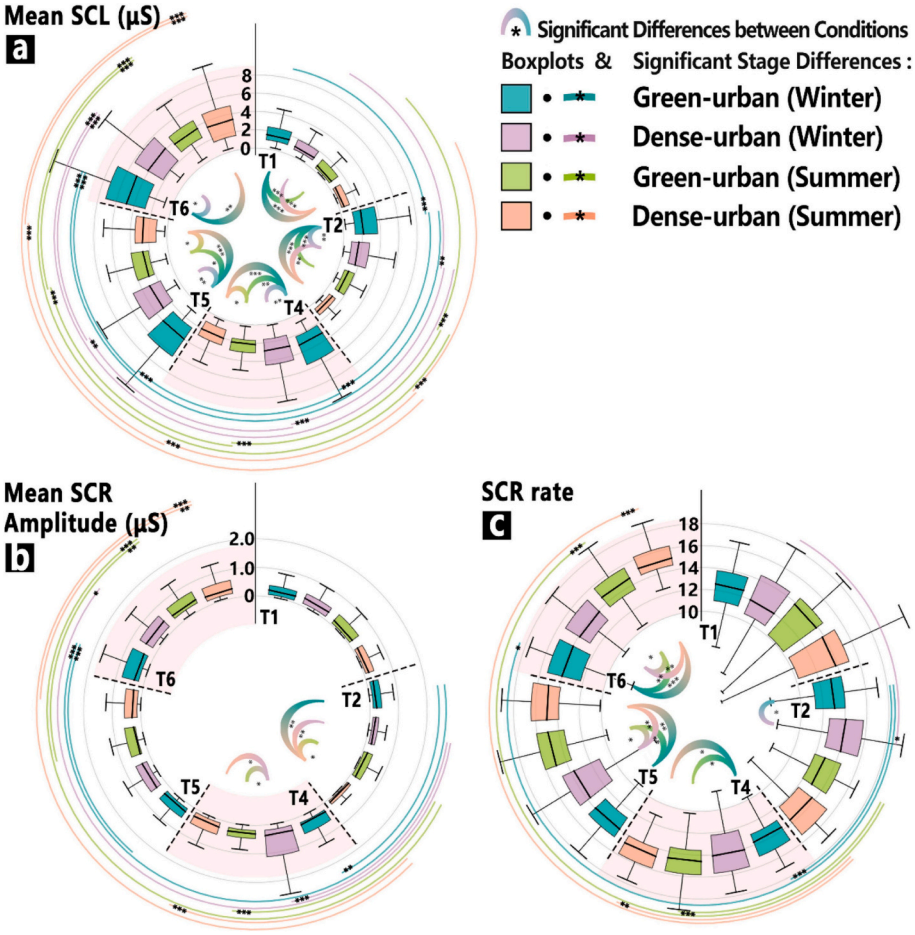


Fig. 6. Boxplots of stage-level restorative responses of physiological measures with Wilcoxon test results across experiments and stages: a. Mean Skin Conductance Level; b. Mean Skin Conductance Response Amplitude; c. Skin Conductance Response Rate. (Note: Thin curved lines outside the boxplots indicate statistically significant pairwise differences between stages within the same experiment. Thick gradient-colored arcs within the central circle indicate statistically significant differences between experiments at the same stage. Connected groups indicate significant pairwise comparisons, whereas the absence of connecting arcs indicates non-significant between-group comparisons. Significance levels are denoted as $p < 0.05$: *, $p < 0.01$: **, $p < 0.001$: ***.)

Table 4
Reliability and convergent validity of the measurement model.

Dependent variable (Latent constructs)	Outer loadings	Composite reliability (CR)	Average variance extracted (AVE)
Positive Subjective Outcomes	–	0.756	0.622
Δ PRS	0.952		
Δ PANAS-Positive	0.580		
Electrodermal Arousal	–	0.743	0.614
Δ SCL	0.512		
Δ SCR Amplitude	0.983		
Perceived environmental quality	–	0.877	0.782
Δ Visual Sat.	0.856		
Δ Acoustic Sat.	0.911		

Note: Δ indicates baseline-corrected values (T6 − T4). Measurement quality criteria were satisfied when outer loadings were > 0.708, composite reliability values ranged from 0.70 to 0.90, and AVE values exceeded 0.50, indicating adequate indicator reliability, internal consistency reliability, and convergent validity.

satisfactory (Hair Jr et al., 2021). Indicators with loadings between 0.40 and 0.708, such as PANAS-Positive and ΔSCL (0.580 and 0.512, respectively, Table 4), were retained if internal consistency and convergent validity were not compromised.

Internal consistency reliability was evaluated using *composite reliability* (CR), with values between 0.70 and 0.90 indicating satisfactory to good reliability (Hair Jr et al., 2021). Convergent validity was assessed via *average variance extracted* (AVE), with a minimum acceptable threshold of 0.50, meaning the construct accounts for at least 50% of its indicators' variance (Hair Jr et al., 2021). In this study, *composite reliability* ranged from 0.743 to 0.877 and *average variance extracted* from 0.614 to 0.782, confirming acceptable internal consistency and convergent validity (Table 4).

Discriminant validity of reflective constructs was assessed using the *heterotrait-monotrait ratio* (HTMT), which compares each construct with all others. Values below 0.85 indicate satisfactory discriminant validity (Henseler et al., 2015). The observed *heterotrait-monotrait ratio* values ranged from 0.039 to 0.302, suggesting that all constructs met the discriminant validity criteria (see Appendix B Table B.1).

3.3.2. Structural model assessment

Path coefficients (β) and their statistical significance (p-values), coefficients of determination (R^2), and effect sizes (f^2) were calculated to evaluate the structural relationships. The initial model included control variables (age, BMI, gender, and PRND) with direct paths to all endogenous constructs. Most controls were non-significant. Only PRND showed a significant path to Negative Affect ($p = 0.018$), and BMI showed marginally significant paths to Positive Subjective Outcomes and Negative Affect ($p = 0.072$ and 0.079, respectively). Non-significant paths were removed from the final model. The path coefficients among the constructs are summarized in Fig. 7, with full details in Appendix B, Table B.2, including bootstrapped path coefficients, 95% confidence intervals, and p-values.

R^2 values indicated the model explained 23.1% of variance in Skin Temperature, 19.7% in Positive Subjective Outcomes, 15.8% in Perceived Environmental Quality, and 12.1% in Electrodermal Arousal, with lower proportions for Thermal Satisfaction ($R^2 = 0.078$), Negative Affect ($R^2 = 0.079$), and SCR Rate ($R^2 = 0.034$). Effect sizes (f^2) quantified the relative contribution of each exogenous construct to the explained variance of endogenous constructs. Following guidelines, $f^2 =$ 0.02, 0.15, and 0.35 indicate weak, moderate, and strong effects, while values below 0.02 are negligible (Cohen, 2013, Hair et al., 2012). As shown in Table 5, most significant predictors had weak effects, suggesting limited but non-negligible contributions to the explained variance.

3.3.3. Pathways from physical environment to restorative responses

After model specification, path coefficients (β), p-values, and effect sizes (f^2) were jointly considered to evaluate both statistical and practical contributions to variance explained (Fig. 7 and Table 5).

Walking environment type (green-urban vs. dense-urban) was significantly associated with Positive Subjective Outcomes ($\beta = -0.605$, $p < 0.001$), with a small-to-moderate effect ($f^2 = 0.110$). This indicates that environment type accounted for a meaningful portion of the variance in perceived restorativeness and positive affect. Ventilation

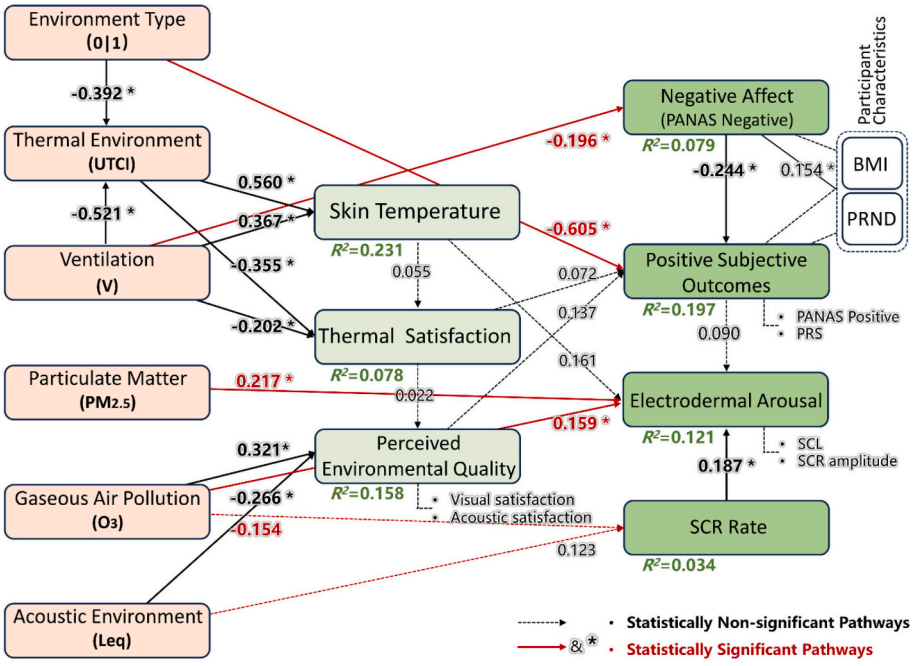


Fig. 7. Results of the PLS-SEM structural model. (Note: Solid lines with asterisks denote statistically significant standardized path coefficients, whereas dashed lines represent non-significant paths. PRND: Participant's residential neighborhood density.)

Table 5
Effect sizes ($f^2 \geq 0.02$) for predictors, mediators, and covariates in the PLS-SEM model.

Predictor / Covariate	Dependent Variable (DV)	*f*-square
Physical Environmental Conditions	**Restorative responses**	
Environment Type	Positive Subjective Outcomes	0.110*
Ventilation (*V*)	Negative Affect	0.041*
Gaseous Air Pollution (O_3)	SCR Rate	0.024*
Gaseous Air Pollution (O_3)	Electrodermal Arousal	0.026*
Particulate Matter ($PM_{2.5}$)	Electrodermal Arousal	0.049*
Physical Environmental Conditions	**Potential Mediation Indicators**	
Thermal Environment (UTCI)	Skin Temperature	0.294**
Ventilation (*V*)	Skin Temperature	0.127*
Gaseous Air Pollution (O_3)	Perceived Environmental Quality	0.114*
Acoustic Environment (Leq)	Perceived Environmental Quality	0.080*
Thermal Environment (UTCI)	Thermal Satisfaction	0.076*
Ventilation (*V*)	Thermal Satisfaction	0.028*
Potential Mediation Indicators	**Restorative responses**	
Perceived Environmental Quality	Positive Subjective Outcomes	0.022*
Skin Temperature (*Tsk*)	Electrodermal Arousal	0.029*
Covariates	**Restorative responses**	
BMI	Negative Affect	0.025*
BMI	Positive Subjective Outcomes	0.024*
PRND	Negative Affect	0.025*

Note: Effect sizes (f^2) quantify the relative contribution of each exogenous construct (predictor or covariate) to the explained variance of the endogenous construct (dependent variable), reflecting effect magnitude only as a variance-based measure and not indicating directionality. PRND: participant's residential neighborhood density. Effect sizes: $f^2 < 0.02$: no effect; 0.02–0.15: * (weak); 0.15–0.35: ** (moderate); ≥0.35: *** (strong).

conditions (*V*) during walking were significantly linked to Negative Affect ($\beta = -0.196$, $p = 0.013$), although the effect size was small ($f^2 = 0.041$), suggesting only modest contribution to explained variance.

For physiological measures, $PM_{2.5}$ and gaseous air pollution (O_3) showed significant effects on Electrodermal Arousal ($\beta = 0.217$, $p = 0.007$; $\beta = 0.159$, $p = 0.029$, respectively), with weak but detectable effect sizes ($PM_{2.5}$: $f^2 = 0.049$; O_3: $f^2 = 0.026$), indicating statistically significant yet practically modest contributions. Marginal associations were observed between SCR rate and O_3 or acoustic environment (Leq) ($\beta = -0.154$, $p = 0.050$; $\beta = 0.123$, $p = 0.052$), with corresponding f^2 values of 0.024 and 0.015, suggesting minimal incremental explanatory value. Consistent with the Wilcoxon results that SCR rates were generally higher during dense-urban walking, these findings imply that part of SCR activity likely reflected traffic-related environmental influences. Therefore, SCR rates cannot be interpreted solely as a physiological arousal indicator linked to restorative responses.

Overall, although several physical environmental variables reached statistical significance, their effect sizes remained small, implying that these predictors primarily provided incremental improvements to model R^2 rather than serving as dominant explanatory drivers of restorative responses.

3.3.4. Assessment of endogenous constructs and potential mediation pathways

Potential mediation pathways were examined by evaluating path coefficients (β) and effect sizes (f^2) between physical environmental variables and the three endogenous constructs.

The acoustic environment (Leq) and O_3 were significantly associated with Perceived Environmental Quality ($\beta = -0.266$, $p = 0.002$; $\beta = 0.321$, $p < 0.001$, respectively), with weak effect sizes for variance explained (Leq: $f^2 = 0.080$; O_3: $f^2 = 0.114$), indicating limited incremental influence on overall R^2. Thermal environment (UTCI) and ventilation (*V*) were strongly linked to Skin Temperature ($\beta = 0.560$, $p < 0.001$; $\beta = 0.367$, $p < 0.001$, respectively), with moderate to weak effect sizes (UTCI: $f^2 = 0.294$; *V*: $f^2 = 0.127$). Their effects on Thermal Satisfaction were weaker (UTCI: $\beta = -0.355$, $p = 0.001$, $f^2 = 0.076$; *V*: $\beta = -0.202$, $p = 0.028$, $f^2 = 0.028$), suggesting only small contributions to explained variance.

Despite small effect sizes from Perceived Environmental Quality to Positive Subjective Outcomes ($f^2 = 0.022$) and from Skin Temperature to Electrodermal Arousal ($f^2 = 0.029$), no statistically significant paths were observed among the endogenous constructs or to restorative responses (Fig. 7). Consistent with results of bootstrapped indirect effects used to assess mediation (Appendix B, Table B.3), all indirect pathways linking physical environmental conditions to restorative responses were non-significant. In addition, two-tailed bootstrapping revealed no significant direct association between Positive Subjective Outcomes and Electrodermal Arousal ($\beta = 0.090$, $p = 0.377$, $f^2 = 0.009$).

3.4. Linear mixed-effects models analysis

The structure and performance of the LMMs for restorative responses, including both best-fitting and alternative models, are presented in Table 6. However, due to singular matrix issues in the best-fitting models based on PLS-SEM for PRS and SCR counts, these results were ultimately omitted. The stepwise results of the LMMs, including the baseline model AIC and R^2 as well as changes in model performance, are provided in Appendix C Tables C.1. Season-adjusted sensitivity analyses (Appendix C Tables C.2-C.3) generally improved model fit while preserving the overall pattern of environmental associations. The independent O_3 effect was attenuated after adjustment for season, consistent with the pronounced seasonal variation of ambient O_3 driven by photochemical formation (Orellano et al., 2020). The remaining environmental associations were largely preserved, suggesting that the influence of seasonal adjustment was largely confined to the O_3 association.

In the PANAS-Positive models, visual and acoustic satisfaction were included stepwise for the modest contribution of Perceived Environmental Quality in the PLS-SEM model ($f^2 = 0.022$), with visual satisfaction (Visual Sat.) showing superior performance. For Electrodermal Arousal, skin temperature ($f^2 = 0.029$), $PM_{2.5}$ ($f^2 = 0.049$), and O_3 ($f^2 = 0.026$) were sequentially added to SCL and SCR amplitude models. Skin temperature, as a physiologically adaptive measure partially regulated by individual thermoregulatory processes, was treated as a continuous covariate and not applied for environment × predictor interaction analyses.

3.4.1. Subjective restorative responses

Including Visual Sat. increased the baseline R^2 for PANAS-Positive from 0.107 to 0.186. Visual Sat. had a significant favorable effect ($\beta = 0.258$, $p < 0.001$), whereas covariates were not significant. The Dense-urban walking × Visual Sat. interaction was non-significant ($p = 0.672$), and adding it did not improve model fit (R^2 and AIC remaining largely unchanged). This indicates that the positive effect of visual satisfaction occurs regardless of environment type.

For PANAS-Negative, adding ventilation (*V*, $f^2 = 0.041$) slightly increased R^2 from 0.132 to 0.138. Wind velocity had a significant inverse effect ($\beta = -0.075$, $p = 0.045$). The Dense-urban walking × *V* interaction was non-significant ($p = 0.166$) and did not improve model fit, with R^2 and AIC both increasing slightly, suggesting that ventilation reduces negative affect independently of environment. Although walking duration was marginally non-significant ($\beta = -0.065$, $p = 0.056$), the trend aligns with stage-level Wilcoxon results, indicating potential attenuation of negative affect with longer walking within the range of this study.

3.4.2. Physiological restorative responses

For SCL, adding O_3 and skin temperature increased R^2 from 0.091 to 0.116. Walking duration positively affected SCL ($\beta = 0.252$, $p < 0.001$),

Table 6
Linear mixed-effects models structure and results for restorative responses.

Restorative responses/ Fixed Effect	β (95% CI)	β (95% CI)
PANAS-Positive	Best-fitting model (R^2: 0.186, AIC: 606.011)	Alternative model (R^2: 0.187, AIC: 607.831)
Dense-urban walking	−0.036 (−0.190, 0.119)	−0.027 (−0.187, 0.133)
Gender	0.302 (−0.325, 0.929)	0.301 (−0.326, 0.928)
Age	0.200 (−0.056, 0.455)	0.200 (−0.055, 0.455)
BMI	−0.032 (−0.320, 0.255)	−0.033 (−0.320, 0.255)
PRND	0.086 (−0.158, 0.330)	0.087 (−0.158, 0.331)
Walking Duration	0.015 (−0.051, 0.081)	0.014 (−0.052, 0.080)
Visual Sat.	**0.258 (0.177, 0.340)** ***	**0.287 (0.132, 0.441)** ***
Dense-urban walking × Visual Sat.	–	−0.036 (−0.201, 0.130)
PANAS-Negative	Best-fitting model (R^2: 0.138, AIC: 621.134)	Alternative model (R^2: 0.141, AIC: 621.212)
Dense-urban walking	−0.042 (−0.195, 0.112)	−0.055 (−0.208, 0.099)
Gender	0.673 (−0.025, 1.372)	0.674 (−0.023, 1.371)
Age	0.209 (−0.075, 0.494)	0.207 (−0.077, 0.491)
BMI	−0.204 (−0.524, 0.116)	−0.206 (−0.525, 0.114)
PRND	−0.091 (−0.363, 0.182)	−0.093 (−0.365, 0.178)
Walking Duration	−0.065 (−0.131, 0.001)	−0.063 (−0.129, 0.003)
V	**−0.075 (−0.147, −0.002)** *	**−0.152 (−0.283, −0.021)** *
Dense-urban walking × *V*	–	0.106 (−0.044, 0.256)
SCL	Best-fitting model (R^2: 0.117, AIC: 629.852)	Alternative model (R^2: 0.118, AIC: 631.238)
Dense-urban walking	−0.139 (−0.306, 0.027)	−0.133 (−0.300, 0.035)
Gender	−0.294 (−1.002, 0.414)	−0.294 (−1.003, 0.415)
Age	−0.099 (−0.388, 0.190)	−0.100 (−0.389, 0.189)
BMI	0.104 (−0.221, 0.430)	0.105 (−0.221, 0.431)
PRND	0.052 (−0.224, 0.329)	0.051 (−0.226, 0.327)
Walking Duration	**0.252 (0.180, 0.324)** ***	**0.252 (0.181, 0.324)** ***
Tsk	0.079 (−0.007, 0.166)	0.082 (−0.004, 0.169)
O_3	**−0.161 (−0.240, −0.082)** ***	**−0.204 (−0.336, −0.071)** **
Dense-urban walking × O_3	–	0.062 (−0.092, 0.215)
SCR Amplitude	Best-fitting model (R^2: 0.053, AIC: 678.027)	Alternative model (R^2: 0.054, AIC: 679.854)
Dense-urban walking	0.121 (−0.072, 0.314)	0.115 (−0.080, 0.310)
Gender	−0.311 (−1.008, 0.386)	−0.312 (−1.008, 0.385)
Age	−0.015 (−0.299, 0.269)	−0.014 (−0.298, 0.270)
BMI	0.058 (−0.262, 0.378)	0.058 (−0.262, 0.378)
PRND	0.061 (−0.211, 0.333)	0.062 (−0.209, 0.334)
Walking Duration	**0.152 (0.072, 0.232)** ***	**0.151 (0.071, 0.231)** ***
Tsk	0.085 (−0.010, 0.181)	0.083 (−0.013, 0.179)
$PM_{2.5}$	**−0.243 (−0.457, −0.029)** *	**−0.235 (−0.452, −0.018)** *
O_3	−0.067 (−0.154, 0.021)	−0.041 (−0.189, 0.107)
Dense-urban walking × $PM_{2.5}$	**0.285 (0.057, 0.514)** *	**0.275 (0.042, 0.508)** *
Dense-urban walking × O_3	–	−0.037 (−0.209, 0.136)

Note: PRND: participant's residential neighborhood density. Significance levels are indicated as follows: $p < 0.05$: *, $p < 0.01$: **, and $p < 0.001$: ***.

while O_3 had a significant inverse effect ($\beta = -0.161$, $p < 0.001$). This indicates that walking elevates SCL, but gaseous air pollution suppresses it. The Dense-urban walking × O_3 interaction was non-significant ($p = 0.434$) and did not improve model fit. R^2 remained similar, and AIC increased slightly, suggesting that the effect of O_3 is independent of environment type.

For SCR amplitude, inclusion of *Tsk*, O_3, $PM_{2.5}$, and the Dense-urban walking × $PM_{2.5}$ interaction increased marginal R^2 from 0.036 to 0.053 and decreased AIC from 679.825 to 678.027. Walking duration increased SCR amplitude ($\beta = 0.152$, $p < 0.001$), while $PM_{2.5}$ had an inverse effect ($\beta = -0.243$, $p = 0.027$). The Dense-urban walking × $PM_{2.5}$ interaction was significant ($p = 0.015$), indicating that the effect of $PM_{2.5}$ on SCR amplitude differed by environment context, with a more pronounced effect observed in the dense-urban walking environment. The Dense-urban walking × O_3 interaction was non-significant ($p = 0.677$) and did not improve model fit (R^2 remaining similar and AIC increasing). Additionally, directional differences between the PLS-SEM pathways and subsequent LMM estimates were observed for associations between physiological restorative responses and air pollution indicators (O_3, and $PM_{2.5}$), and are further discussed in the Discussion section.

To further examine the interactive effects among physical environmental exposures during walking, additional interaction analyses were conducted based on the PLS-SEM findings. As only $PM_{2.5}$ and O_3 showed significant pathways to electrodermal arousal (Fig. 7), an interaction term between these two pollutants was further incorporated into the best-fitting LMMs for SCL and SCR amplitude. As shown in Table 7, the $PM_{2.5} \times O_3$ interaction term showed a marginally significant association only in the SCR amplitude model ($p = 0.070$), whereas no statistically significant effect was observed for SCL. In addition, sensitivity analyses replacing O_3 with NO_2 (Appendix C Table C.4) showed higher explanatory power than the primary O_3-based models, while the overall pattern of associations remained highly consistent. O_3 was retained in the primary analyses based on the predefined model selection strategy. The comparable findings indicate that NO_2 can also serve as a reasonable alternative indicator of gaseous air pollution.

3.4.3. *Integrated interpretation of LMMs*

LMM results corroborated the PLS-SEM and stage-based analyses. Individual-level covariates were generally non-significant, highlighting that environmental and perceptual factors drove observed effects. For subjective measures, visual satisfaction was the strongest predictor of positive affect across environments. Ventilation modestly reduced negative affect regardless of environmental context. In physiological measures, walking duration consistently elevated tonic and phasic electrodermal arousal. Gaseous air pollution (O_3) suppressed SCL independently of the environment. Particulate matter ($PM_{2.5}$) negatively affected SCR amplitude, with a significantly stronger effect observed in dense-urban settings. However, no clear interaction effects among physical environmental exposures were observed in the present study.

4. Discussion

Using a randomized crossover design across two seasons, this study initially demonstrated that nature-based walking produced higher perceived restorativeness than dense-urban walking, consistent with prior evidence (Chen et al., 2025, Ma et al., 2024, Tu et al., 2026b). In

Table 7
Linear mixed-effects models for potential interactions among physical environmental indicators.

Restorative responses/ Fixed Effect	β (95% CI) **SCL** (R^2: 0.118, AIC: 631.202)	β (95% CI) **SCR Amplitude** (R^2: 0.059, AIC: 676.775)
Dense-urban walking	−0.139 (−0.306, 0.027)	0.104 (−0.088, 0.297)
Gender	−0.291 (−1.000, 0.418)	−0.309 (−1.009, 0.391)
Age	−0.100 (−0.389, 0.189)	−0.019 (−0.304, 0.266)
BMI	0.103 (−0.223, 0.429)	0.057 (−0.265, 0.379)
PRND	0.049 (−0.228, 0.326)	0.053 (−0.220, 0.327)
Walking Duration	0.250 (0.178, 0.322) ***	0.148 (0.068, 0.228) ***
Tsk	0.076 (−0.010, 0.163)	0.074 (−0.022, 0.169)
O_3	−0.150 (−0.233, −0.068) ***	−0.034 (−0.128, 0.059)
$PM_{2.5}$	–	−0.194 (−0.413, 0.024) †
Dense-urban walking × $PM_{2.5}$	–	0.256 (0.027, 0.485) *
$PM_{2.5} \times O_3$	0.042 (−0.061, 0.145)	0.109 (−0.009, 0.227) †

Note: PRND: participant's residential neighborhood density. Significance levels are indicated as follows: $p < 0.10$: †, $p < 0.05$: *, $p < 0.01$: **, and $p < 0.001$: ***.

contrast, regardless of environment type, negative affect decreased across all walking conditions, indicating a general mood-regulating effect of walking itself. This pattern was supported by LMM results showing a marginal inverse association between walking duration and negative affect ($\beta = -0.065$, $p = 0.056$), aligning with meta-analytic findings on walking-related mental health benefits (Hanson and Jones, 2015; Xu et al., 2024).

Within the green-urban condition, lower temperatures were associated with stronger restorative responses. Although seasonal differences at T6 were negligible, winter green-urban walking showed enhanced restorativeness. This was reflected in higher PRS and PANAS-Positive scores, as well as relatively elevated SCL. This pattern is consistent with prior findings from the same region indicating diminished visual restorative benefits when air temperature exceeds 33 °C (Dong et al., 2025), supporting a moderating role of thermal conditions. Although summer temperatures in this study were moderate (28.6 ± 2.7 °C) and below the 33 °C threshold, winter green-urban walking still showed slightly stronger restorative effects, consistent with earlier trends.

4.1. Multiple exposure-response pathways influencing restorative responses

During urban walking, restorative responses appeared to arise through multiple parallel pathways. Rather than reflecting a unified recovery process, affective and physiological responses showed partially distinct associations with perceived environmental quality and physical environmental stressors.

Affective restoration was primarily associated with visual satisfaction and environmental perception. Although walking environment type significantly influenced perceived restorativeness and positive affect in the PLS-SEM analysis, the LMM results further indicated that visual satisfaction was strongly associated with positive affect across both walking contexts. This pattern suggests that the emotional benefits of natural settings may depend on how the environment is perceptually appraised during walking rather than on environment type per se. Environments perceived as pleasant may reduce emotional strain and facilitate short-term affective restoration, broadly consistent with Stress Recovery Theory (Ulrich et al., 1991). Ventilation also modestly reduced negative affect, possibly because improved airflow influences thermal sensation and heat storage during walking (Zhang et al., 2020), thereby alleviating prolonged environmental discomfort. Together, these findings suggest that affective restoration may be closely linked to perceptual and sensory experience.

In contrast to the affective pathways, PLS-SEM did not identify significant associations between subjective restorative measures and air pollution exposure, whereas physiological pathways showed significant associations with air pollution exposure. Across both analytical approaches, $PM_{2.5}$ and O_3 were associated with electrodermal responses, while LMM indicated reduced tonic skin conductance and SCR amplitude. This suggests a direct influence of air pollution on autonomic arousal regulation, and aligns with evidence that traffic-related air pollutants can affect physiological stress regulation during pedestrian activity (Cavalcante de Sá et al., 2016; Du et al., 2022). One possible explanation is that, given the relatively short exposure duration and moderate pollution levels in the present study, pollutant exposure during walking may not always be consciously perceived or sufficiently integrated into stable perceptual experience. In parallel, EDA, as an index of sympathetic nervous system activation, remains sensitive to subtle environmental stressors (Kreibig, 2010) and reflects concurrent physiological responses that may not be reported by subjective measures.

Collectively, these findings suggest that affective and physiological restoration may operate as partially distinct regulatory processes within urban walking environments, broadly corresponding to the complementary yet not fully integrated relationship between Attention Restoration Theory (ART) and Stress Recovery Theory (SRT) (Joye et al., 2024). ART primarily emphasizes cognitive and affective processes involved in attentional recovery (Kaplan and Kaplan, 1989; Kaplan, 1995), whereas SRT focuses on the reduction of psychophysiological stress, including autonomic stress regulation and associated negative affect (Ulrich, 1983, Ulrich et al., 1991). Although both processes may be concurrently activated by the same environmental exposure, their responses are not necessarily coupled in magnitude or timing, as they may involve multiple restorative mechanisms and regulatory dynamics.

4.2. Interaction and independent effects of environmental exposure during walking

The effect of environmental context on exposure-response relationships was generally limited, with most identified local environmental exposures independently eliciting measurable restorative responses, subsequently confirmed through LMM analyses.

Among the exposure variables identified by the PLS-SEM, only one significant interaction between urban walking and $PM_{2.5}$ was detected in the LMM analysis. In particular, dense-urban walking × $PM_{2.5}$ was significant ($\beta = 0.285$, $p = 0.015$) for SCR amplitude, suggesting a stronger $PM_{2.5}$-related suppression of phasic EDA in dense-urban settings. Comparable findings have been reported in a virtual reality experiment, where natural environments produced stronger restorative outcomes and lower physiological stress indicators (e.g., SCL) than urban settings under equivalent noise exposure (Kawai et al., 2024). Together, these results suggest that, under comparable mild or localized adverse environmental exposures, nature-based settings may directly mitigate autonomic stress responses to specific factors.

Nevertheless, no other significant environment interactions were detected for the other exposure-response pathways identified by the PLS-SEM. Independent effects emerged along specific dimensions: O_3 modulated physiological responses (e.g., SCL and SCR amplitude), whereas perceptual and experiential factors (e.g., perceived visual quality and ventilation) influenced affective responses independently of environmental context. These findings suggest that the majority of restorative effects from environmental exposure operate largely regardless of the surrounding context.

Correspondingly, winter dense-urban walking produced limited short-term restorative effects in this study. Specifically, SCR amplitude was significantly higher during the first 1 km in winter than in summer, accompanied by improvements in perceived restorativeness and positive affect. These observations complement the pathway analysis, indicating that restorative responses are not exclusive to natural environments, but can also occur in built settings under favorable climatic conditions. Similar effects have been reported in historic urban areas with limited natural elements (Stigsdotter et al., 2017) or specific built scenes (Collado and Manrique, 2020). However, these effects diminished at 2 km, suggesting that such benefits are transient, season-dependent, and unlikely to persist without supportive environmental contexts.

The observed independent effects of major environmental exposures reveal that restorative responses are shaped not only by environmental context but also by the combination of exposures, highlighting that increasing green space alone may be insufficient. These findings provide empirical support for managing key physical environmental factors to enhance restorative potential in built urban areas with limited opportunities for greening.

4.3. Temporal scale differences in environmental exposure effects

Environmental effects on restorative responses operate over different temporal scales. Although winter green-urban conditions produced measurable restorative benefits, short-term variations in thermal indicators (UTCI, skin temperature, thermal satisfaction) did not show significant pathways to subjective or physiological responses in the PLS-SEM analysis. Stage-level thermal fluctuations were small (ΔT4–T6: 0.853–1.250 °C), below typical perceptual thresholds for discomfort

(Battistel et al., 2023; Du et al., 2014). Thus, micro-scale variations were insufficient to trigger noticeable psycho-physiological changes.

The lack of stage-level thermal effects may also reflect a mismatch in temporal scale. Thermal comfort depends on ~35 s of exposure history (Xie et al., 2022) and short-term thermal sensation is based on ~2–3 min accumulated experience (Lau et al., 2019). The stage aggregation window (15 min) likely missed both immediate fluctuations and the 15–20 min needed for steady-state subjective responses (Goto et al., 2006, Greco et al., 2016). Still, stronger responses in winter green-urban conditions suggest thermal effects operate at broader contextual or seasonal scales, not micro-scale stage fluctuations. In contrast, air pollution stressors ($PM_{2.5}$ and O_3) showed short-term, even immediate, pedestrian-level effects on physiological responses.

Likewise, variations across stages in acoustic indicators (Leq and acoustic satisfaction) were not associated with significant restorative response pathways in the PLS-SEM analysis (Leq on SCR: $p = 0.052$; all other pathways were non-significant). Both walking environments were characterized by moderate-to-high noise exposure (Leq: green-urban 58.37 ± 1.81 dBA; dense-urban 66.52 ± 1.68 dBA). Although the green-urban route exhibited comparatively lower noise levels, this difference may have been insufficient to produce distinct restorative effects within the model. Consistent with recent findings, perceptual evaluations in moderate-noise (≈60 dBA) and high-noise (≈70 dBA) settings tend to resemble each other more closely than those in low-noise environments (≈50 dBA) (Lu et al., 2020), suggesting limited perceptual differentiation under the exposure levels observed in this study.

Additionally, the partial discrepancy in coefficient direction between the PLS-SEM and LMM findings for $PM_{2.5}$ and O_3 associations with electrodermal responses likely reflects differences in exposure representation and temporal aggregation between the two analytical frameworks. The PLS-SEM was based on T6–T4 pairwise differences derived from independent within-participant observations, reflecting middle-to-post walking changes in physical environmental conditions and corresponding intervention-related restorative responses across walking stages. Although the pairwise difference preserved relative exposure–response variation patterns, it may have compressed stage-level exposure variability and attenuated information related to absolute environmental exposure levels. By contrast, the LMM analyses retained absolute exposure conditions together with stage-level within-person restorative response fluctuations. The differing coefficient directions may therefore suggest that electrodermal responses to air pollutants are not strictly linear across exposure ranges, as also reported in previous cross-sectional studies (Di et al., 2017; Hua et al., 2025), such that relative pairwise-difference-based associations and stage-level absolute exposure effects may not necessarily exhibit identical directional patterns. Consequently, the PLS-SEM and LMM may reflect different aspects of the exposure–response relationship, emphasizing relative environmental contrasts within the multivariate restorative-response system and stage-level physiological fluctuations under specific exposure conditions, respectively.

Overall, different environmental exposure effects appear to operate over distinct temporal windows, with pollutants showing short-term physiological associations, whereas thermal conditions are more evident at broader contextual or seasonal scales.

4.4. Strength and limitations

This study investigates the multi-pathway mechanisms underlying restorative walking and offers several key strengths. Conceptually, building upon environmental comparisons and adopting a preliminary exploratory approach, the research treats restorativeness as a system simultaneously shaped by multiple environmental factors. Methodologically, it synchronously captured pedestrian-level environmental exposures and corresponding restorative responses across multiple dimensions within the same participants under equivalent measurement scales, enabling direct comparability. Analytically, parallel exposure-response pathways were modeled using PLS-SEM to explore multi-exposure, multi-pathway mechanisms within a unified framework. These models were subsequently validated using LMMs, which also assessed potential environmental interactions. However, several limitations should be noted.

As a field-based experiment, the results were inevitably influenced by real-world environmental variability, which limited the contrast between conditions. For example, the seasonal sessions were conducted in a fixed, non-counterbalanced order, such that seasonal and session-order confounding could not be fully disentangled. Consistent with this design structure, modest restorative responses were observed even during the first half of winter dense-urban walking, likely due to the presence of street greenery and relatively high environmental quality along the selected routes. Although these factors may have diminished the distinctiveness between dense-urban and green-urban conditions, the exposure-response pathways identified in the analyses remained detectable, indicating that the affective benefits of green-urban walking were still present but less pronounced in comparison.

Methodologically, several limitations arise from the analytical strategy, which is primarily associated with differences in exposure representation and temporal aggregation between modeling frameworks. The discrepancy in the direction of several coefficients is mainly attributable to the application of pairwise differences in PLS-SEM, which may compress variability in exposure levels. In addition, within the PLS-SEM framework, thermal (UTCI) and acoustic (Leq) indicators showed limited variability and no significant pathways to restorative responses, differing from associations reported in previous single-factor studies (Schaupp et al., 2025, Song et al., 2024). This discrepancy may reflect temporal aggregation effects, as thermal and acoustic responses can operate over accumulation windows shorter or longer than the 15-min stage intervals applied here. Future studies using finer temporal segmentation may better capture these scale-sensitive associations.

Several measurement-related limitations should also be acknowledged. Although missing EDA data were limited and randomly distributed, mean imputation may have introduced a conservative bias by reducing physiological variability, thereby attenuating effect sizes and limiting the detection of marginal effects in the PLS-SEM results. To minimize participant burden, the satisfaction assessment was simplified, and some sensory dimensions (e.g., olfaction) were not included. This may have limited the representation of multidimensional environmental perceptions and partly explained the lack of significant mediation effects involving satisfaction variables. Accordingly, the role of integrated perceived environmental quality in restorative processes warrants further investigation.

Finally, limitations related to sample characteristics and external validity should be considered. Healthy young adults were recruited to reduce potential physiological bias, which may be associated with the non-significant sex differences observed across analyses. Furthermore, this sampling strategy limits the generalizability of the findings to other age and health groups. The same walking workload may correspond to different physical intensity and physiological burden in middle-aged and older adults, as well as in populations with clinical conditions, potentially resulting in different restorative responses. Future research could include more diverse population groups to improve external validity.

5. Conclusion

This study confirms that nature-based walking produces stronger short-term restorative responses than dense-urban walking. Notably, the PLS-SEM results indicate that these responses are driven by multiple parallel pathways rather than a single integrated route. Specifically, perceptual and experiential factors predominantly influence affective responses, whereas environmental stressors such as air pollutants act directly on physiological responses (e.g., EDA), with minimal evidence of cross-pathway mediation. These parallel pathways suggest that restorative processes operate through multidimensional mechanisms

and cannot be reduced to a single factor.

The LMM analyses tested a limited number of interaction effects and identified a significant interaction between dense-urban walking and $PM_{2.5}$ on SCR amplitude. This indicates that particulate exposure effects may be amplified under compounded dense-urban environmental stressors. Although contextual moderation was limited overall, this finding suggests that environmental exposures do not operate as simple linear additions, but may interact under certain environmental conditions. Most other exposure-response relationships remained stable and independent across walking environments, providing empirical support for environmental planning and management strategies that target specific physical indicators to enhance restorative potential.

Scale-dependent effects were observed, with thermal conditions operating primarily at broader contextual or seasonal levels, whereas traffic-related pollutants exerted more immediate influences at the pedestrian exposure scale. This differentiation across spatial and temporal scales highlights that environmental stressors should not be treated in isolation. These findings underscore the need to consider perceived quality, micro-environmental exposures, and broader contextual factors together when designing and managing urban walking environments.

Overall, the findings indicate that even within the same walking context, heterogeneous environmental exposures can produce differentiated short-term restorative outcomes. Importantly, this highlights that restorative walking environments arise from integrated exposure-response processes involving both psychological and physiological pathways, rather than being attributed to greenery alone.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Data availability

Data will be made available on request.

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